Mapping Flood Damage from Space — Report

Multimodal AI Hackathon 2026, Track B

1. Problem

When roads, bridges and phone lines are gone, rescuers need to know where the damage is and who is cut off. Our system takes **an area and a flood date** and, from free satellite data only, answers: **(1) where did the flood hit, (2) what is damaged, (3) who is cut off.** It runs end-to-end from raw satellite data for any area and date (nothing is hard-wired for Nepal). Case study: the 26 August 2026 Bhote Koshi–Trishuli flood. This is an educational prototype, not an operational tool.

2. Data (allowed sources only)

Sentinel-1 radar and Sentinel-2 optical images (Copernicus Data Space / Sentinel Hub, AWS Open Data), Copernicus DEM GLO-30, OpenStreetMap **as it was before the event** (ohsome API snapshot at 00:00 UTC the day before; never Overpass or live OSM), and the training datasets Kuro Siwo and Sen1Floods11. **Copernicus EMS, UNOSAT or any published damage map is never an input** — not for training, thresholds or pipeline; EMSR927 appears only in `validation/` and in Section 8 to check finished results. Thresholds are set from training data or literature, never tuned on EMSR927.

3. System overview

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area + date → download (area only): S1 before/after, same orbit track, both pass directions; S2 if clear; DEM; OSM snapshot
 → preprocessing (10 m UTM grid, layover/shadow mask, slope, HAND)
 → flood water (AI model)   → debris / mud / flooded vegetation (change test)
 → final flood map → damage (buildings, roads, bridges, facilities)
 → road network before vs after → cut-off settlements, repair order
 → stats.json → one-page situation report + map overlays → dashboard
```

Steps communicate through files in one run folder; `stats.json` is the single source of every number in the dashboard and report. Key rules: radar change detection uses **pre and post from the same relative orbit**; the “before” image is checked so that it is not already flooded (water-fraction time series, earlier image used if needed); the main track is the earliest post-event image unless the opposite direction sees the valley floor clearly better. If no usable pair exists, the system says *insufficient data*; if nothing is found it says *no significant flooding* **together with coverage** (what could be seen).

4. AI component: flood segmentation model

Model. U-Net with ResNet34 encoder (ImageNet initialisation, allowed by the organisers), 7 input channels: before/after VV and VH backscatter (dB), their differences, and terrain slope; 3 classes (no water / permanent water / flood). Trained on Kuro Siwo (Bountos et al., 2024) plus Sen1Floods11 (Bonafilia et al., 2020) with Lovász loss; probabilities calibrated (temperature 0.92). Sentinel-1 is processed to match Kuro Siwo (sigma0, Lee 7×7 + 3×3 mean): measured bias 0.00 dB, speckle 0.96× of Kuro Siwo on 28 tiles. Runs on a CPU (~30 s per 30×30 km per track). An optional optical U-Net (ResNet18, Sentinel-2) adds water where the sky is clear. Post-processing: no water on slopes > 15° or HAND > 30 m, radar shadow checked from both directions, patches < 0.1 ha removed; *Standard* and *Sensitive* modes (probability threshold 0.25).

Results (events never seen in training; pixels pooled):

Test	Water class	Precision	Recall	F1	IoU
Kuro Siwo test (10 events, 15 896 tiles)	flood	0.869	0.708	**0.780**	0.639
Kuro Siwo test	permanent water	0.811	0.749	0.779	0.638
Kuro Siwo test	water total	0.895	0.763	0.824	0.700
Kuro Siwo test, mountain tiles (relief > 200 m)	flood	–	–	0.48	–
Sen1Floods11 Bolivia (held out), no “before” image	water total	0.959	0.118	**0.210**	0.118
Sen1Floods11 Bolivia, “before” = “after” (no change)	water total	0.889	0.495	0.636	–

Test	Water class	Precision	Recall	F1	IoU
Optical model, Sen1Floods11 Bolivia (held out)	water	0.899	0.872	**0.885**	0.794

For reference, the best model in the Kuro Siwo paper reaches F1 0.801 on these events. We report the Bolivia numbers **honestly**: the radar model learned to rely on change between before and after, so with no “before” image it rarely reports water (F1 0.21); Bolivian water is also bright in radar (median −16 dB). That setting is not how the pipeline runs (it always has a “before” image), but it shows the model is a **change detector, not a general water detector**. Differences between model variants after round 3 are within seed noise (val score ±0.007); the only clearly repeatable gain came from the terrain-slope input.

Himalayan scenes the model has not seen. No training dataset contains a Himalayan event, so we built a small held-out set with labels from Sentinel-2 only (no EMS): Chamoli 2021, Teesta 2023 and a no-flood control (Marsyangdi 2024). Preliminary results (v0.2, radar from a proxy source): **0 % false flood on dry land in all three**, but recall of narrow, fast mountain rivers is low (Teesta 0.18, Marsyangdi 0.15 with one direction, 0.38 with both; Chamoli 0). Using both pass directions raised usable area by 16 % (Teesta). Final numbers with the real pipeline input: *to be added when the run finishes*.

5. Debris, mud and other surface change

Mud, rock and flooded vegetation/buildings do not look like open water in radar: debris and wet mud often darken only slightly, flooded vegetation and towns **brighten** (double bounce). A separate statistical step compares each pixel with its own history (up to 4 earlier passes of the same track), combines both pass directions, grows regions from strong seeds, and labels subtypes (debris, mud, erosion, flooded vegetation, flooded / damaged buildings, unknown) with a confidence level using Sentinel-2 and OSM building/land-use when available. A **self-test on a flood-free image pair** estimates how much of the result may be false; a weather check (change over the surrounding slopes) rejects rain and wet snow. Measured on the Trishuli valley floor: 6.3 % darkening vs 0.4 % brightening; on Opava 2024: 0.5 % / 13.6 %. Tested on a synthetic valley (all subtypes ≥ 90 %, no false pixels elsewhere); real-data thresholds come from literature on floods and landslides, **not measured on Himalayan debris flows**. Results are therefore shown as indicative, with confidence.

6. Damage

Per object, not per village: a building is *affected* if its outline plus a 10 m margin overlaps the flood map, *likely destroyed* if debris/erosion lies under it, *possibly affected* if only the margin or low confidence. Roads are *blocked* at ≥ 17 % of length or ≥ 30 m continuous in the flood zone (stricter rules cut false closures from 16 % to 5 %, Kasmalkar 2020). Bridges: water under a bridge is not evidence of damage; approaches and debris are tested → *blocked / suspect / passable*. Counts are given as a range (certain only … including “possible”), like UNOSAT’s “potentially affected”. No population estimates (only OSM `population` if present).

7. Who is cut off

We build our own road graph from the pre-event OSM snapshot (networkx; OSMnx would use live OSM). Post-event graph = pre-event graph without blocked road pieces and blocked bridges; a settlement is *connected*, *degraded* (detour > 2×), *possibly cut off* (cut only if suspect bridges/possible blockages are counted), *cut off*, or *no road before the event*. Destinations: hospitals, towns and the edge of the downloaded network. Output includes detour %, walking time, blocking road pieces, a five-level impact grade per settlement with reasons, an urgency ranking, and a **stepwise repair plan** (which road section reconnects the most buildings per repaired section). Helicopter landing-zone candidates from slope in the DEM are always marked *unverified*. Bonus: a click on the map traces the flood path downstream (D8 flow direction on the DEM) and lists settlements, bridges and facilities along it, ordered downstream — no hydraulics, no arrival times.

8. Case study: Trishuli flood vs EMSR927

Results to be inserted from the validation module. EMS data: “European Union, Copernicus Emergency Management Service data”.

9. Limitations (what the system cannot do)

- **Revisit time.** Sentinel-1/2 pass an area only every few days (about 6–12 days per track). The system maps what was there on the image dates; it **cannot warn of a sudden event such as a glacier collapse minutes ahead** and may miss short-lived water. A very recent flood may have no “after” image yet; the app then says so and when a better view is expected.

- **Radar shadow and layover** in steep valleys hide part of the ground. Two pass directions help but do not cover everything; the data-quality panel states what share was visible.
- **Narrow, fast mountain rivers** look like land to radar (rough water, channel 2–6 pixels wide); Himalayan recall is low.
- **Clouds** (monsoon) limit optical use; then only radar is used.
- **Debris vs wet soil.** Debris/mud detection is heuristic, with literature thresholds not measured on Himalayan debris flows; C-band does not penetrate dense forest; streets aligned with the radar look direction hide urban flooding.
- **The model is a change detector** (see Bolivia result) and is weaker on mountain terrain than on plains (F1 0.48 vs 0.78). Differences between variants are within noise.
- **OpenStreetMap is incomplete** in remote areas (roads, bridges, buildings missing) and its pre-event state is all we know. Footpaths and footbridges are not part of the road network, so a village reachable only by a path is “no road before the event”.
- **10 m pixels:** a “hit” building means flood on or beside it, not a damage grade; no depth or flow velocity.
- **No population estimates**; landing-zone candidates are from the DEM only (trees, wires and obstacles are unknown).
- **No ground truth** was used to tune thresholds; reference comparison only at the end (Section 8).
- **“No significant flooding”** means nothing was found in the available data, not that nothing happened.
- Needs free accounts (Copernicus, ohsome) and internet; the 30×30 km area limit is a time and quota choice.

10. Responsibility

The disaster is recent and many families are still affected. We use no images of victims. Results are a first screening from satellite data, unverified on the ground, and must not be the sole basis of rescue decisions.

11. AI assistance

Code, documentation and this report were developed with **Claude Code** (Anthropic); the team reviewed and can explain every part. Details per module are in the README (section *AI assistance*).

12. Attributions and citations

Contains modified Copernicus Sentinel data 2026. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors (ODbL). Training data: Bountos, N. I. et al. (2024), Kuro Siwo, NeurIPS Datasets and Benchmarks; Bonafilia, D. et al. (2020), Sen1Floods11, CVPR Workshops. Validation only: European Union, Copernicus Emergency Management Service data (EMSR927).